Product Introduction

The ZDH1333 is a wideband 50Ω RF power amplifier operating from 50MHz to 2GHz, characterized by high linearity and low noise. It features internal dynamic bias circuitry to overcome the adverse effects of temperature variations. The ZDH1333 is suitable for operation from +5V to 8V. Furthermore, it consists of two independent amplifiers arranged in a push-pull configuration, exhibiting excellent second-harmonic characteristics.

The ZDH1333 is manufactured using a gallium arsenide (GaAs) pHEMT process and packaged in a QFN3x3-16 lead-free green package. The ZDH1333 achieves performance levels comparable to similar international products, offering excellent value for money.

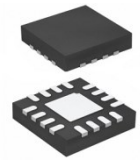
Typical application scenarios

- DOCSIS 3.x Applications
- Distribution Amplifiers
- CATV Line Driver Amplifier
- FTTB/FTTH Output Amplifier

Absolute Maximum Ratings

Parameter	Value
Storage temperature	-65°C~+150°C
Operating temperature	-55°C~+125°C
Supply voltage (VDD)	+10V
RF input power	+15dBm

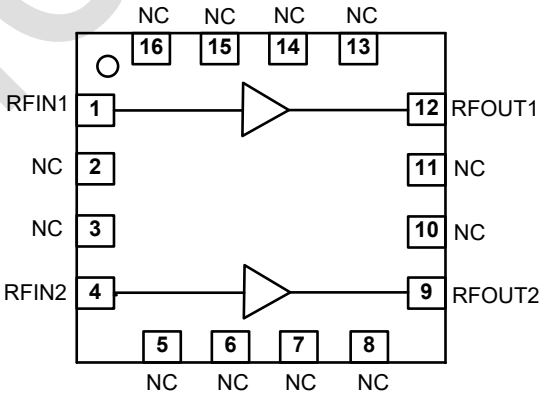
Features



- 5V or 8V single-voltage supply, typical current:
  230mA @5V, 375mA @8V
- Typical gain: 20dB @ 500MHz, VDD=5V;
  20dB @ 500MHz, VDD=8V
- Typical OIP3: 40dBm @ 500MHz, VDD=5V;
  43dBm @ 500MHz, VDD=8V
- Typical P1dB: 26dBm @ 500MHz, VDD=5V;
  30dBm @ 500MHz, VDD=8V
- Input/output 50Ω impedance matching
- Green lead-free 16-pin QFN3x3 package

Pb This product complies with all relevant regulations and is halogen-free.

Pin diagram (Top View)



Pin No.	Name	Description
1	RFIN1	RF Input 1
4	RFIN2	RF Input 2
9	RFOUT2	RF Output 2
12	RFOUT1	RF Output 1
Other	NC	Floating or grounded

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Electrical parameters

1. Test conditions: VDD=+5 V, Iq=230mA, Temp=+25° C, 50MHz~2GHz application circuit, 50Ω test system.

Parameter	Typical value					Unit
Frequency	50	500	900	1450	2000	MHz
Gain	20	20	19	16	15	dB
Input Return Loss (S11)	-10	-20	-18	-8	-7	dB
Output Return Loss (S22)	-13	-34	-12	-9	-15	dB
Noise Figure (NF)	1.6	1.8	1.9	2.7	3.4	dB
Output Power 1dB Gain Compression Point (P1dB)	25	26	24	21	21	dBm
Third-Order Intermodulation(1) (OIP3)	39	40	37	32	32	dBm

Note: (1) The two tones are spaced 1MHz apart, and the output power of each tone is 5dBm.

2. Test conditions: VDD=+8 V, Iq=375mA, Temp=+25°C, 50MHz~2GHz application circuit, 50Ω test system.

Parameter	Typical value					Unit
Frequency	50	500	900	1450	2000	MHz
Gain	20	20	19	17	16	dB
Input Return Loss (S11)	-11	-22	-17	-8	-8	dB
Output Return Loss (S22)	-13	-33	-12	-10	-15	dB
Noise Figure (NF)	1.8	1.9	2.0	2.8	3.6	dB
Output Power 1dB Gain Compression Point (P1dB)	30	30	29	25	25	dBm
Third-Order Intermodulation(1) (OIP3)	40	43	40	37	36	dBm

Note: (1) The two tones are spaced 1MHz apart, and the output power of each tone is 5dBm.

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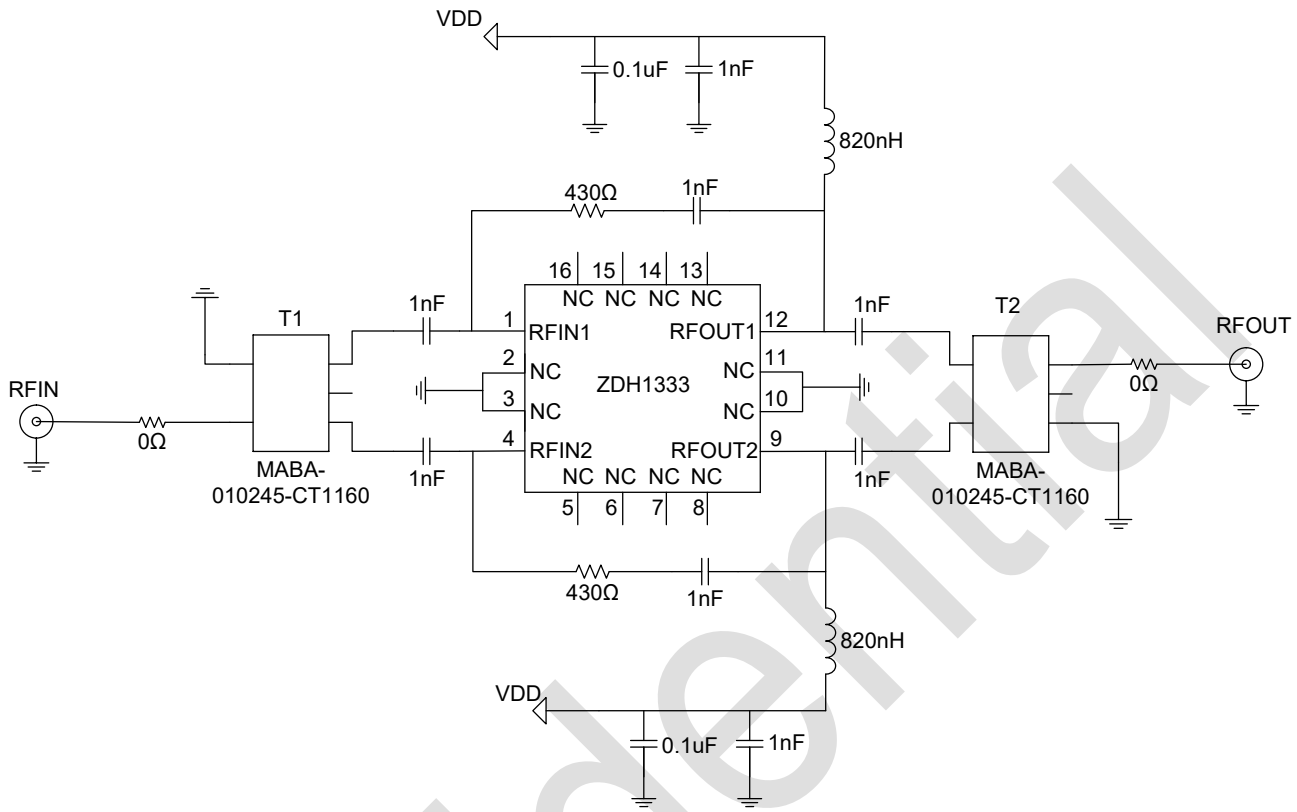
Application circuit diagram (50MHz~2000MHz)



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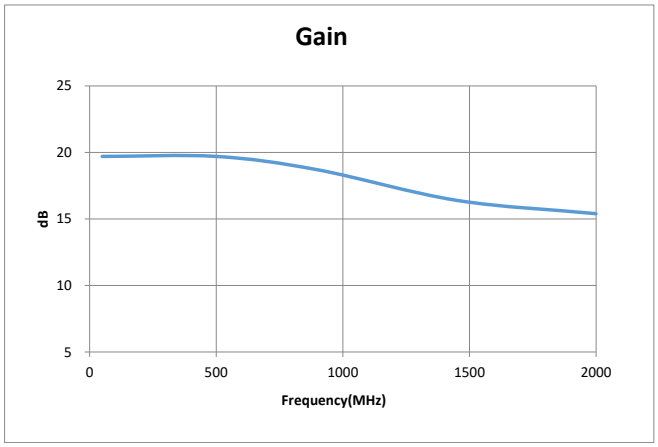
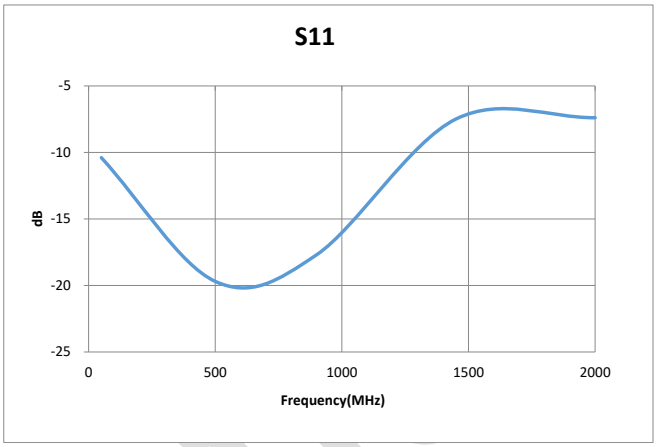
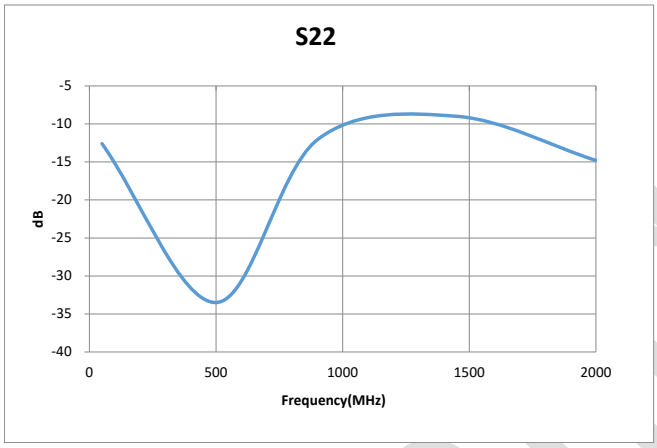
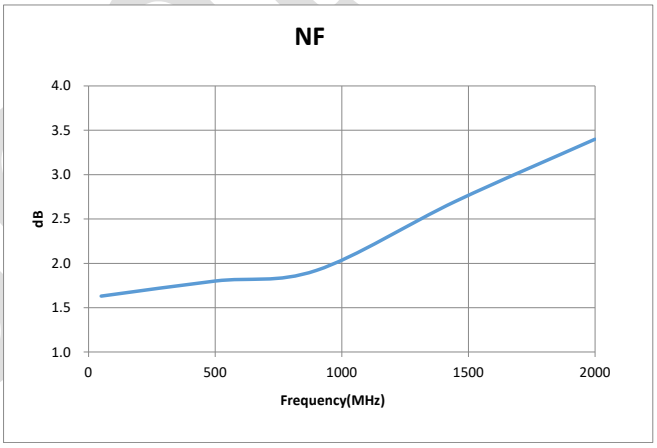
Typical performance curves

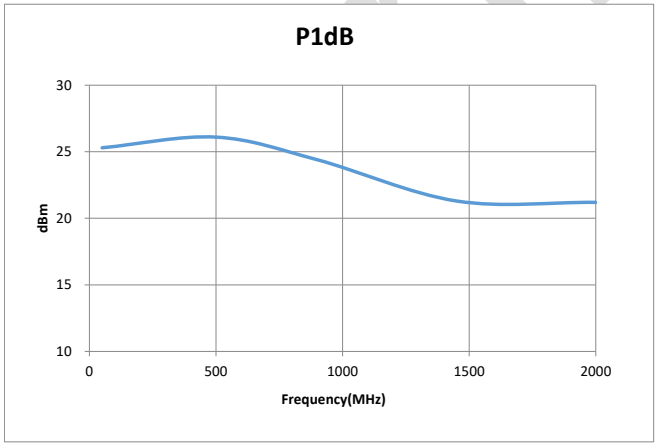
1. Test conditions: VDD=+5 V, Iq=230mA, Temp=+25°C, 50MHz~2GHz application circuit, 50Ω test system.

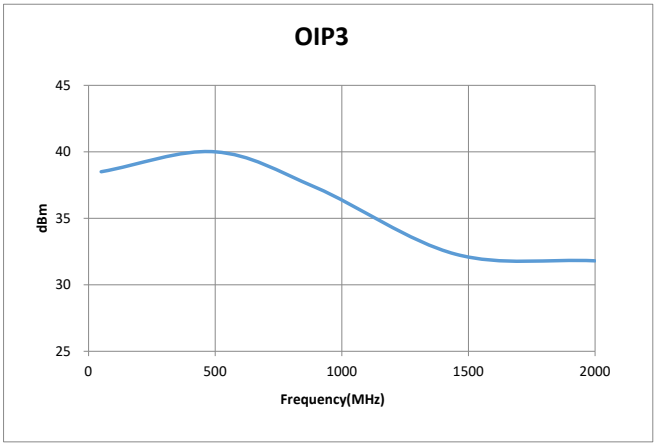






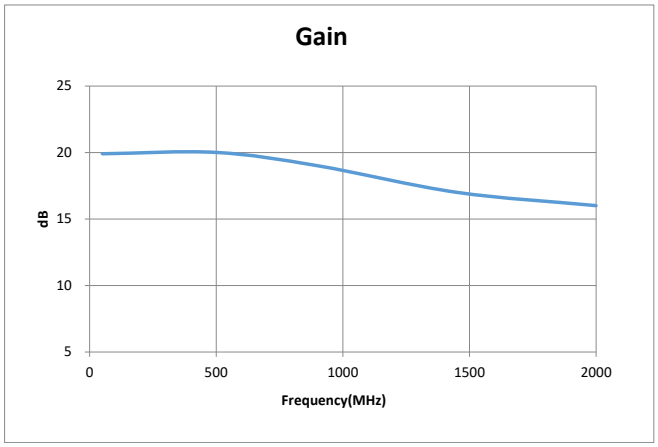


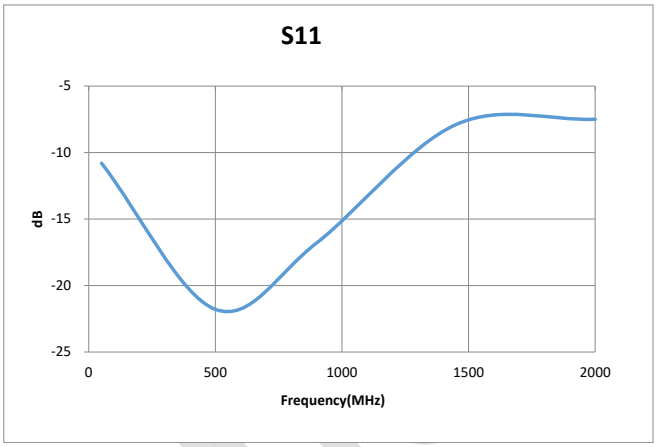


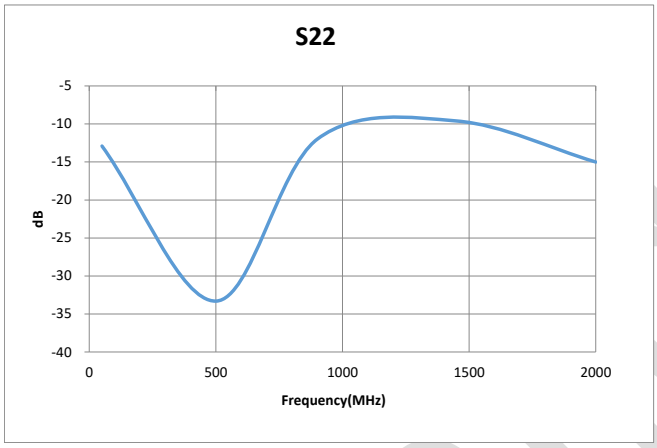


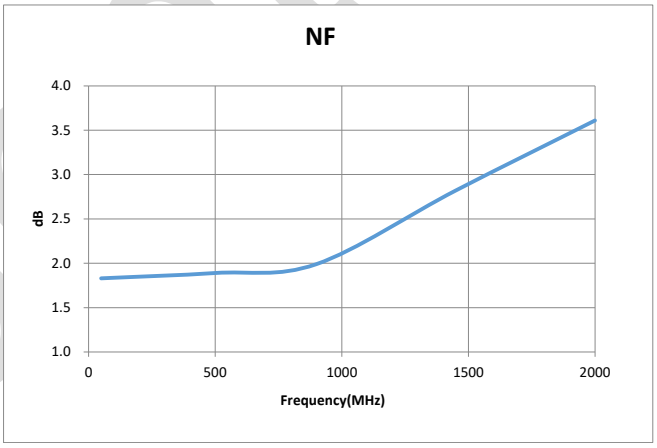
Typical performance curves

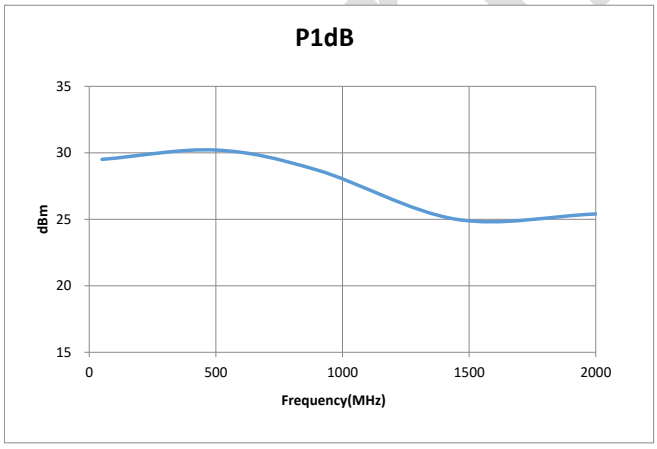
2. Test conditions: VDD=+8 V, Iq=375mA, Temp=+25°C, 50MHz~2GHz application circuit, 50Ω test system.

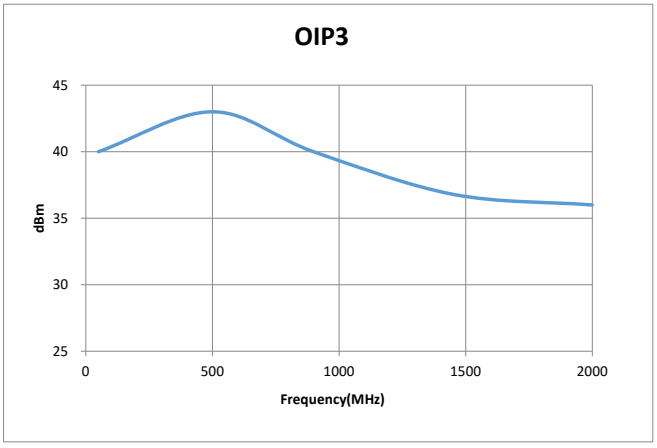




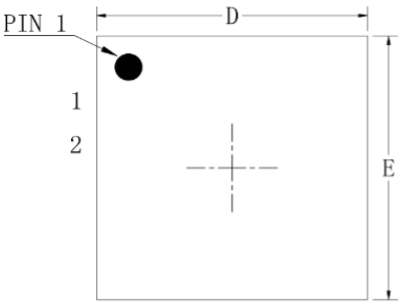




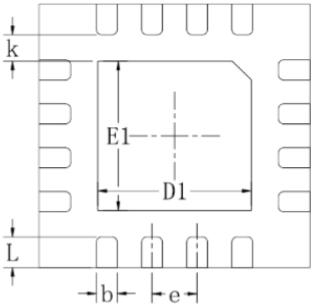




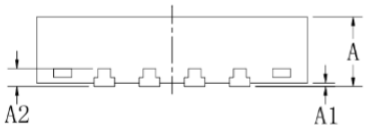
Package size diagram



top view



bottom view



side view

symbol	Dimensions unit: mm		
	minimum value	Standard value	maximum value
A	0.700	0.750	0.800
A1	0.000	--	0.050
A2	0.195	0.203	0.211
D	2.950	3.000	3.050
E	2.950	3.000	3.050
D1	1.600	1.700	1.800
E1	1.600	1.700	1.800
k	0.300 Min.		
b	0.180	0.230	0.280
e	0.500 Typ.		
L	0.300	0.350	0.400

Ordering Information

Model	Marking	Packaging
ZDH1333	ZD1333	QFN3x3-16

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